Research Methods

How to summarise an Article/Paper

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http://depts.washington.edu/psywc/handouts.shtml (not accessible at the moment)

Content

- What is a research article?
- How to read an article
- Writing the Summary
- Summarising and Plagiarism
- Conclusion
- References

Before we start

- What is an article or a paper?
- How do you read an article?
- What are the main problems facing you in understanding an article?
- What are the main sections used in an article?
- When you will be ready to summarise an article?
- What do you write in the summary?

What is a research article/Paper?

- It uses a standard format to clearly communicate information about an experiment
- A research article usually has Eight major sections:
  - Title
  - Abstract
  - Introduction and background- the WHY
  - Literature Survey- the WHAT
  - Method used- the HOW
  - Experiments & Results
  - Discussion, conclusion and future work
  - and References

Article section boundaries

- Sometimes there are minor variations, such as:
  - a combined sections including
    - Results and Discussion, or introduction & Literature review
  - or an overall General Discussion section includes
    - multiple experiments presented in one article.
  - A finishing section as the conclusion and future work

- Sections might vary in different ways, but the main sections are there all the time

How to read an article

How to read an article (1..4)

- **Enough Time to read**; allow yourself reasonably enough time, this takes more time than most students realise

- **Understand**; before you can write about the research, you have to understand it
  - Then you need to answer the following:
    - Does the author's study make sense to you in lay terms?
    - Could you explain the study to your peers?

- **Explain**; when you can clearly explain the study in your own words, then you are ready to write about it

- Here's how to proceed

How to read an article (2..4)

1. Scan the article first
   - If you try to read a new article from start to finish, you'll get bogged down in details
2. Briefly look at each section to identify the following:

What to identify	Section
The research question and reason for the study	Introduction
Hypothesis or hypotheses tested	Introduction
How the hypothesis was tested	Method
The findings	Method
Results, including tables and figures	Experiment(s)
How the findings were interpreted	Discussion

How to read an article (3..4)

3. Underline/highlight key sentences or write the key point of each paragraph in the margin *(these are the building blocks of your summary)*

4. Although the **abstract** can help you to identify the main points, you cannot rely on it exclusively, because it contains highly condensed information

5. Read for depth, read interactively

How to read an article (4..4)

6. After you have highlighted the main points, read each section several times

7. As you read, ask yourself these questions:

- *How does the design of the study address the question posed?*
- *What are the controls for each experiment?*
- *How convincing are the results?*
- *Are any of the results surprising?*
- *How does this study contribute toward answering the original question?*
- *What aspects of the original question remain unanswered?*

Writing the Summary

Writing the Summary (1..6)

- Like an abstract in a published research article, the purpose of an article summary is to give the reader a brief, structured overview of the study

- To write a good summary, identify Why, What, How
  - what relevant information is important and condense that information for your reader
- ***The better you understand a subject, the easier it is to explain it thoroughly and briefly***

- **Writing the first draft- STORY!**
  - Use **the same order** as in the article itself
  - **Adjust the length** accordingly depending on the content of the particular article

Writing the Summary (2..6)

...Writing the first draft

What to write	Sentences
State the research question and explain why it is interesting	1
State the hypothesis/hypotheses tested	1
Briefly describe the methods design, procedure, what was manipulated, what was measured, how data were analysed.	1-3
Describe the results, What differences were significant?	1-3
Explain the key implications of the results, avoid overstating the importance of the findings	1

The results, and the interpretation of the results, should relate directly to the hypothesis

Writing the Summary (3..6)

... **Writing the first draft**

- The number of suggested sentences given is only a rough guideline for the relative length of each section.
- For the first draft, focus on content, not length (it will probably be too long), condense later as needed.
- Try writing about the hypotheses, methods and results first, then about the introduction and discussion last.
- If you have trouble on one section, leave it for a while and try another one.

Writing the Summary (4..6)

... **Writing the first draft**

- Edit for completeness and accuracy
- Add information for completeness where necessary
- More commonly, if you understand the article, you will need to cut redundant or less important information
- Stay focused on the research question, be concise, and avoid generalities
- The Methods summary is often the most difficult part to edit

Writing the Summary (5..6)

Finalising your summary

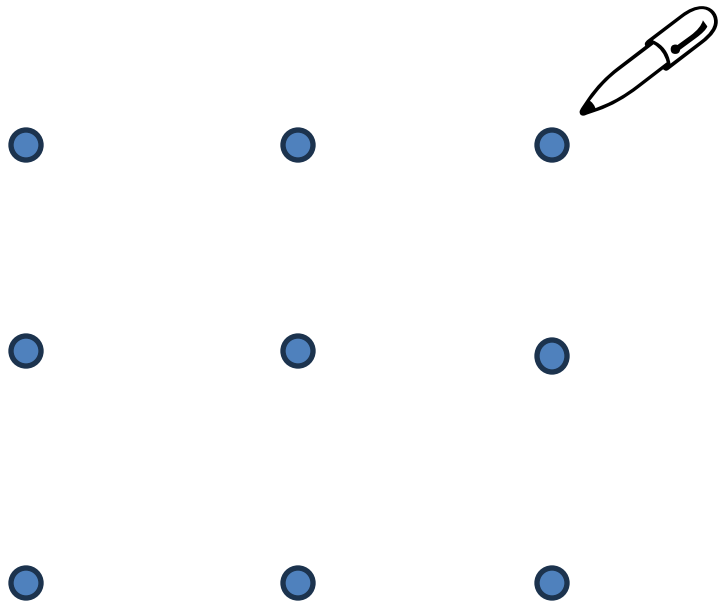
- Edit for style. Write to an intelligent, interested, naive, and slightly lazy audience (e.g., yourself, your classmates).

- Expect your readers to be interested, but don't make them struggle to understand you.

- Include all the important details; don't assume that they are already understood.
  - Eliminate wordiness, including most adverbs ("very", "clearly").
    - "~~The results clearly showed that~~ there was no difference between the groups" can be shortened to
    - "There was no significant difference between the groups".
  - Use specific, concrete language.
  - Use precise language and cite specific examples to support assertions.
  - Avoid vague references (e.g. "this illustrates" should be "these result illustrates").

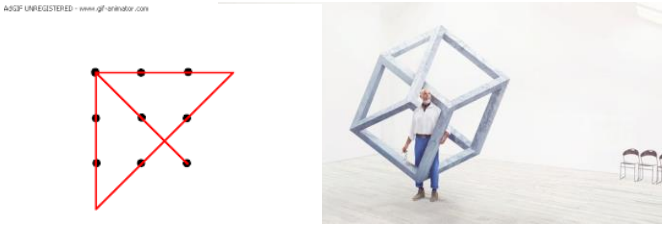
Try this exercise

Use a pen to draw only FOUR lines visiting all the nine points without lifting the pen from the paper.



Writing the Summary (6..6)

... Finalising your summary (go outside the box)



- Use scientifically accurate language.
  - For example, you cannot "prove" hypotheses (especially with just one study). You "support" or "fail to find support for" them.
- Rely primarily on rephrasing, not direct quotes.
- Direct quotes are seldom used in scientific writing. Instead, rephrasing what you have read.
- To give due credit for information that you rephrase, cite the author's last name and the year of the study (Smith, 1982). (e.g. “APA Citations”)
- Re-read what you have written.
- Ask others to read it to catch things that you have missed.

Summarising, Plagiarism, and Conclusion

Summarising and Plagiarism

- Plagiarism is always a risk when summarising someone else's work. To avoid it:
  - Take notes in your own words
  - Avoid writing complete sentences when note-taking
  - Summarise points in your own words
  - If you find yourself sticking closely to the original language and making only minor changes to the wording, **then you probably don't understand the study** (see the university handout, about plagiarism)

Conclusion

Understand the way of summarising an article has more than one benefit:

- **First**; it will be easy for you to draw comparisons between more than one relevant articles
- **Second**; it is an accumulated knowledge/skills helps to plan your road map toward the progression of your research
- **Third**; if you practice summarising articles, it will be easy for you to write your own publication after knowing the main elements that are used to construct an article
- **Fourth**; Summarising different related articles, group them together, and critically analyse them is the building block of your research literature review

References

- Summarizing a Research Article, University of Washington, Psychology writing center 2004, *This link is broken at the moment* http://depts.washington.edu/psywc/handouts.shtml

- Project in computing and information systems, a student's guide, second edition, by Christian W. Dawson, Addison Wesley 2009, ISBN 978-0-273-72131-4